Final referee report

A general entropy-tail law for adaptive rare-event estimation
Consolidated mathematical audit | 29 October 2026

Recommendation. Retain the principal results and revise the exposition before acceptance. The audits found no counterexample to Theorem 3.2, the stated finite-cap program comparison, the optimized genie convergence theorem, or the fixed-cap identification threshold. The most important revision is to make the finite-divergence qualification of pointwise genie convergence unmistakable.

Correction to the earlier audit characterization. The unconditional statement that the genie ratio converges for every admissible quadruple is false. However, the supplied Theorem G.2 already introduces its convergence estimate with "If $C_v < \infty$". Its concluding sentence naturally inherits that hypothesis. Accordingly, this report does *not* classify the conditional theorem as disproved. The counterexample establishes a sharp scope limitation and motivates an explicit qualification, rather than a retraction of the main theorem.

Finding category	Disposition
Actual mathematical error	Unconditional pointwise genie convergence is false. No unambiguously unconditional assertion of it was located in the supplied source; do not attribute the false extension to the conditional theorem.
Proof clarifications	Make the genie quantifiers and order of limits explicit; preserve midpoint work, common-pair quantifiers, conditional normalization, same-unit coupling, and finite-prefix tail control.
Optional improvements	Ordinary-band mass ≤ ρ instead of 3ρ; ordinary contribution ≤ 2Ψ instead of 9Ψ; an explicit optimizer for Ψ; optional constant sharpening.
Main theorem	The full stated law/rarity/budget scope survived the audit. One conservative pair of comparison constants is 10^{-17} and 33000.
Scope of assurance	Analytic referee assessment plus independent finite checks. This is not proof-assistant verification, a publication-priority assessment, or a claim of sharp constants.

Manuscript examined. The supplied 51-page standalone PDF and its corresponding review-source ZIP. Section and source-line references below refer to that unchanged version. Proposed patches are supplied separately and have not been applied to the manuscript. Earlier research packages in the workspace are not substituted for this version.

Audit coverage. Section 3/Appendix A; weighted sign entropy and the information gate in Appendix B; common pairs, midpoint accounting, variational obstruction and additive localization in Appendix C; Sections 4.1-4.5; stopping-flow/dual programs and the factor-sixteen guarantee; Appendix G limits; Appendix H identification; and the end-to-end theorem assembly.

1. The genie counterexample and its exact scope

Location: Appendix G.2, Theorem G.2; sections/genie_brackets.tex, lines 99-111. The sentence "In particular both brackets converge to K_∞" should explicitly retain the preceding finite-divergence condition. The optimized statement in Theorem G.3 is different and remains valid.

$$\mu(dd) = 2\,dd \text{ on } (0, 1/2], \quad \rho = 1/4,$$
$$q_L = 0, \quad q_H = q_w = 1/4, \quad q_v(d) = 1/2 - d.$$

All four priors are admissible: the variance prior has μ-integral 1/4, and the endpoint target gap is 1/4. With $\delta = q_H - q_L = 1/4$,

$$C_v = (1/8) \int_0^{1/2} [1/(1/4 - d^2)]\,dd = \infty.$$

The policy that always reaches depth N and receives the latent reveal has infinite signed-likelihood information and deterministic work N. Hence $K_N^+ = \infty$ for every finite N. This is not merely a failure of a residual formula to be finite: the actual genie ratio is infinite.

The ordinary ratios remain uniformly bounded. Let p denote the latent Bernoulli mean and let U be the auxiliary uniform law on (0,1). Under q_v, p has density 2(1-p). Under q_w, its density is 3/2 below 1/2 and 1/2 above 1/2. The signed endpoint difference has density -1/2 below 1/2 and +1/2 above it.

For an ordered prefix s of length $n \geq 1$ containing k heads, put $u_s = \int_0^1 p^k(1-p)^{n-k}\,dp$. Then

$$|p_H(s) - p_L(s)| \leq u_s/2,$$
$$p_v(s) = [2(n-k+1)/(n+2)]u_s,$$
$$r_s \leq [(n+2)/(8(n-k+1))]u_s \leq (3n/8)u_s.$$

Summing over stopped prefixes and the independent seed gives information $\leq (3/8)E_U T$. The work-reference density dominates U/2, so information $\leq (3/4)E_{q_w} T$ for every capped randomized policy. The auxiliary U is used only for domination; it need not belong to the rarity class.

$$9/128 \leq K_\infty \leq 3/4, \quad \text{while} \quad K_N^+ = \infty \text{ for every finite N.}$$

Classification. This refutes unconditional pointwise convergence, including the tempting replacement of $C_v < \infty$ by $K_\infty < \infty$. It does not refute the finite-C_v statement, the monotone inequalities, or convergence after optimization. Earlier wording suggesting a disproved conditional theorem should be corrected.

2. Precise replacement text for Appendix G

Replace the statement of Theorem G.2 with the following. Preserve its existing equation labels so references continue to resolve. The companion file referee_genie_scope.patch implements this replacement and inserts the preceding counterexample as a remark.

"For every admissible quadruple of priors, with information ratios interpreted in the extended nonnegative reals,

$$K_N \uparrow K_\infty \le K_N^+, \quad K_{N+1}^+ \le K_N^+.$$

If, in addition, $C_v < \infty$, then

$$0 \le K_N^+ - K_N \le e_N^v, \quad e_N^v \to 0.$$

Consequently $K_N^+ \downarrow K_\infty$ under this finite-divergence hypothesis. Without it, the monotone brackets remain valid, but pointwise convergence of the upper bracket need not hold."

Keep Theorem G.3 unchanged in scope. Its proof already regularizes the variance reference in exactly the way needed to bypass the counterexample. After the invocation of Theorem G.2, insert:

"Keep $\varepsilon > 0$ fixed throughout this limit in N; no convergence of the unregularized quadruple is used."

After sending ε to zero, add: "The limits are taken in this order; no convergence uniform in ε is asserted." The reason is the following complete chain.

$$m = (q_L + q_H)/2,$$
$$q_v^\varepsilon = (1-\varepsilon)q_v + \varepsilon m,$$
$$K_\infty(q_v^\varepsilon) \le K_\infty(q_v)/(1-\varepsilon).$$

Convexity preserves admissibility. Since $|\delta| \le 2m$ and $|\delta| \le 2(1-m)$, both branches of the regularized latent score have absolute value at most $2/\varepsilon$. Thus its latent divergence is finite. At each fixed ε, the conditional pointwise theorem applies.

$$\lim_{N\to\infty} G_N(B)^2 \ge (\Delta\theta)^2 / [4 + BK_\infty/(1-\varepsilon)].$$

Only now let ε decrease to zero and take an optimizing quadruple from Lemma G.1. This proves $G_N \uparrow P_\infty$ without any uniform pointwise genie limit. In the counterexample, for every fixed N, the always-reveal lower bound C_v^ε/N diverges as ε decreases to zero, demonstrating why reversing the limits is invalid.

No downstream theorem needs to be weakened. The optimized bracket $G_N/8 \le R \le P_N$, eventual stopping criterion, and identification results remain supported. The distributional Theorem 3.2 uses Sections 3-4 and Appendices A-C; it does not depend on Appendix G.

3. Proof clarifications: lower-bound interfaces

The following are targeted exposition changes, not additional false statements found in the supplied proof. Several protections are already present; retain them when shortening the manuscript. Suggested insertions make their role easier to check.

B.3: information gate. Keep the finite-variance hypothesis and distinguish V, the uniform variance, from M_0, the null second moment. After the activation construction, add: “Seed-only activation occurs before the first observation, and restart independence is conditional on the fixed latent state. The gate does not reveal a magnitude or select a population component.” This prevents an invalid unconditional independence interpretation.

C.4: a pair common to all policies. Around the finite-intersection argument (transfer_foundations.tex, approximately lines 268-285), add: “For fixed b and δ, the resulting pair is chosen before the adaptive population algorithm and before its deterministic total-observation cap. It is the same pair for every local policy and every cap.” The finite-dimensional compact convex image, finite-list minimax, and weak-star finite-intersection steps are all needed; a policy-dependent pair is insufficient.

C.2 and C.6: midpoint work and unbounded stopping. After the work accounting, add: “The charged expectation is under the midpoint population, which is admissible by convexity. It is not replaced by the average endpoint work.” A tested adaptive example has endpoint works about 2048 but midpoint work about 2.93×10^{10}. Endpoint-average substitution would therefore be false.

At the cap-removal step, make the quantitative justification explicit: “Use the common pair at each cap L. The endpoint transcript coupling error is bounded by $P_-(T > L) + P_+(T > L)$, which tends to zero under finite endpoint expected work. This passage does not assume a common deterministic stopping bound.”

C.6 to Section 4: conditional normalization. Display the following identity immediately before the unnormalized annular formula, with $\theta = \tau r$, $\Gamma_\sigma^\circ = \tau\Gamma^\circ$ and $w_{*,\sigma} = \tau w_*$:

$$\tau^2 \min\{r\Gamma^\circ, w_*/a^2\} = \min\{\theta\Gamma_\sigma^\circ, \tau w_{*,\sigma}/a^2\}.$$

The second branch contains the extra τ. Do not replace it by $w_{*,\sigma}/a^2$. In Section 4.4 explicitly recall that low bands have largest endpoint at most 1/128, parity gives endpoint ratio at least four, and trimming supplies saturation before invoking the annular proposition.

C.5 and Sections 4.1-4.3. Preserve the subtraction 4b. The one-flip example described on page 7 shows that every smaller fixed coefficient can fail. Preserve “inner finite objective, before its outer cap by ρ” in the optimality lemma. State that allocations defining Ψ may split atoms fractionally; hard assignments for entropy selection do not impose a hard-assignment restriction on Ψ.

Section 4.5: all-budget assembly. Add one sentence before factoring the component sum: “The ratio θ_i/G_i is common, including after the fixed bias and annular constants are retained; hence all components have the same budget crossover.” There is no loss depending on the number of components.

4. Proof clarifications: upper bounds and programs

Section 3 and Appendix A: observability. Preserve the full-law null denominator in the likelihood bank. Add after its definition: “The split $\sigma + \tau + \zeta = \mu$ specifies the reference bank and the work analysis; membership in these measures is never observed.” Replacing the denominator by a selected-component null mixture can destroy the false-positive bound.

Same-unit corrections. After the adjacent-level correction definition, add: “Both classifiers act on successive unused observation blocks of the same latent unit. Their marginal error bounds survive the first stopping time by conditional independence of the unused suffix.” If independent fresh units were used, even perfect classifiers would have difference second moment $2\theta(1-\theta)$, instead of a quantity tending to zero with classification error.

Poisson accounting. Retain $E\Delta^2$, not $\mathrm{Var}(\Delta)$, in the compound-Poisson variance. Explicitly state that the seed may choose zero replications, the first observation of every replicated unit is charged, and Poisson means below one are allowed. These facts are what extend the construction to every $B > 0$.

Section 3 completion and Section 4.1 limit. Insert: “Approximation is taken at fixed (μ,ρ,B); the universal comparison constant does not depend on the chosen decomposition, terminal level, or finite prefix. No common optimizer across prefixes is required.” In the upper proof, first take the risk infimum over finite constructions. In the lower proof, retain the omitted tail probability in the prefix comparison. Neither side requires a finite full-law inverse moment.

Finite convex program and factor sixteen. The stopping-flow representation and exact dual survived independent checks. Keep the four priors distinct, the closed-perspective conventions at null states, and the root charge. Next to the factor-sixteen corollary, add: “The comparison holds for the stated sufficient cap N_*, or for a cap passing the verified genie criterion; it is not asserted for every fixed cap.”

$$L \le G_N \le P_\infty \le P_N \le U,$$
$$L/8 \le R \le U, \quad U \le 2L \Rightarrow U \le 16R.$$

A concrete fixed-cap counterexample is $\mu = (\delta_{1/2} + \delta_{1/4})/2$, $\rho = 1/8$, $B = 300000$. It has $P_1/R > 16.65$, whereas the stated sufficient cap is much larger and the cap-one genie test fails. This is not a counterexample to the supplied corollary.

Appendix G.1: compactness. Add that the finite-cap ratios are upper semicontinuous even at infinite denominators, because the numerator is continuous and bounded and the ratio is defined as zero there. The decreasing superlevel sets are compact and nested. This argument proves the ordinary optimized interchange; it does not prove convergence of every genie ratio.

Appendix H: identification. Keep the conclusion about the fixed-cap program floor separate from unrestricted consistency. The threshold is $N \ge 2m-1$ for m distinct positive magnitudes. For infinite support every fixed cap has a positive floor, but caps may grow as B grows. Clarify that identifying the target functional does not require identifying the entire allocation q.

5. Optional improvements, separate from corrections

O1. Sharper ordinary-mass argument, Sections 4.2-4.3. The bound $w_j \le 3\rho$ can be replaced by $w_j \le \rho$. With $x = a_j$, $A = \sum a_i$ and $F = G(a_l)$, use $\log(1+t) \le \sqrt{t}$ to get

$$\begin{aligned}\Delta &= x + A\log(1+x/A) + x\log(1+A/x)\\ &\le x + 2\sqrt{(Ax)} \le x + 2\sqrt{(Fx)}.\end{aligned}$$

Thus $\sqrt{(F+\Delta)} - \sqrt{F} \le \sqrt{x}$. Inner optimality gives $xb_j \le \sqrt{(\rho x)}$, hence $w_j = xb_j^{\,2} \le \rho$. The $F = 0$ case gives the same conclusion directly.

The dyadic top-mass factor also improves from three to two. Select the largest inverse gaps up to mass s. If r is selected at the boundary value f, the selected integral is at least (2s-r)f, while the excluded integral is at most (2s-r)f. Therefore total integral is at most twice the selected integral.

$$hH_0 \le 2\Psi(h), \quad \text{replacing} \quad hH_0 \le 9\Psi(h).$$

The optional patch referee_optional_ordinary_constants.patch changes the mass lemma, its proof, the dyadic lemma and its application, and both occurrences of 10Ψ in the final assembly to 3Ψ. These changes must be made together. Factor one is false: twelve equally weighted atoms at 2^{-j-1}, $\rho = 1/4$ and $h = 2^{-12}$ have a genuinely optimal all-ordinary choice with $hH_0/\Psi \approx 1.11787$.

O2. An explicit optimizer for Ψ, Appendix C.5. Let $\alpha > 0$ solve $\int \min\{1, \alpha^2/d^2\}d\mu = \rho$. Then one allocation works at every h:

$$w_*(d) = \min\{1, \alpha^2/d^2\}.$$

The mass equation has a unique solution for $0 < \rho < 1$. If $h \ge \alpha$, this allocation gives $\Psi = \rho$. If $h < \alpha$, the multiplier $h/(2\alpha)$ maximizes the pointwise concave objective after subtraction of the mass price. This supplies an independent optimizer check without nonatomicity. It is optional; the existing proof does not need attainment.

O3. Constants. The weighted sign-entropy proof supports 214 in place of 257, but changing it is not needed for any conclusion and would require consistent downstream arithmetic. The complete main-theorem chain supports the conservative constants 10^{-17} and 33000. These may be recorded in a remark; they do not improve the separate factor-sixteen program guarantee.

O4. Scope examples. Include one example showing that the retained set need not be an interval, and the infinite-moment tail law from page 8. These would explain why the arbitrary-law selection formulation is substantive. Do not strengthen the novelty claim on the basis of this audit: no fresh systematic priority search was performed.

6. Counterexamples to shortcuts, not to stated results

Tempting replacement	Concrete obstruction and required protection
Genie convergence for every quadruple	The uniform-law example on page 2 has $K\infty \le 3/4$ but every finite genie ratio is infinite. Retain finite C_v for pointwise convergence; regularize before optimizing the limit.
Endpoint-average work for midpoint work	The adaptive trigger example has endpoint works near 2048 and midpoint work near 2.93×10^{10}. Use the admissible midpoint population directly.
Disjoint sign-testing windows	A sufficiently long fixed-horizon majority test makes every window event the same event. Only the weighted overlap inequality is valid.
Unsaturated annular entropy lower bound	With masses $1\text{-}2^{-11}$ and 2^{-11} at gaps $1/128$ and 2^{-64}, a centered one-flip estimator has variance at most 1024; the unsaturated claimed RHS exceeds 4.5×10^{14}. Retain $\min\{\rho\Gamma^\circ, w^*/a^2\}$.
Packing without trimming	A genuine optimal 64-band selection allocates entropy exceeding untrimmed saturation by a factor 3.75. Trimming three bands preserves over 94% of entropy and restores saturation.
Bias defect cb with $c < 4$	For masses $1\text{-}\varepsilon$ and ε at gaps $1/2$ and d, $\varepsilon = 2^{-10}$, $\rho = 1/8$, use $Y = X\text{-}b$ with $b = \varepsilon(1\text{-}2d)/4$. Then $VW_0 \le 1/4$ but the modified modulus product diverges as d decreases to zero.
Optimizer chosen after outer cap	For the point mass at gap $1/2$, $\rho = 1/100$, $h = 1/4$, the empty retained set minimizes the outer-capped objective but gives $hH = 1/2 > 9\Psi = 9/100$. Choose an inner minimizer.
Only indicator allocations in Ψ	For the point mass at gap $1/8$, $\rho = 1/8$, $h = 1/4$, the correct value is $\Psi = \rho$; indicator-only allocation has value zero. Fractional allocation at atoms is essential.
Factor sixteen at arbitrary fixed cap	The two-atom example on page 5 gives $P_1/R > 16.65$. Use the prescribed sufficient cap or the verified genie stopping condition.
Dropping the finite-prefix tail	For masses $7/8$ and $1/8$ at gaps $1/2$ and 2^{-100}, $\rho = 1/8$, $B = 2^{80}$, full $K = 1/8$ but the easy-band prefix is about 6.016×10^{-13}. Retain the omitted probability mass.

These examples establish the necessity of particular hypotheses or proof steps. They must not be relabeled as counterexamples to statements that already contain those hypotheses. The current manuscript explicitly avoids the listed shortcuts.

7. End-to-end result and pathological-law tests

The lower assembly is uniform in budget. Put $\lambda^2 = \rho/F$. Packing gives $\sum G_i \ge F/896$ and $\theta_i = \lambda^2 G_i/1024$. Retaining the actual bias and annular constants, additive localization gives

$$R \ge (\sum G_i/128) \times \min\{\lambda^2/65536, \sqrt{c_{ann}}\,\lambda/(32\sqrt{B})\}, \quad c_{ann} = e^{-28}/2000000.$$

Thus $R \ge c_E \min\{\rho, \sqrt{(\rho F/B)}\}$, with $c_E = \sqrt{c_{ann}}/(896\cdot128\cdot32)$. Combining this with $\Psi/8192$, the printed 10Ψ assembly, the high-gap case and $K \le 2\lim k_J$ yields a lower coefficient about 8.01×10^{-17}. The rounded 10^{-17} bound is certified by rational elementary estimates, not floating-point fitting.

The upper assembly is also uniform. The mixed-work sum has respective entropy and ordinary coefficients below 98 and 25207. Including the 128z bias term, the large-z constant estimator, and the outer rarity cap gives the rounded upper coefficient 33000. No label access, law-dependent multiplicative loss, or minimum expected budget is introduced.

$$10^{-17} K_{\mu,\rho}(B) \le R_{\mu,\rho}(B) \le 33000 K_{\mu,\rho}(B).$$

Infinite inverse moments. Let $\mu\{2^{-j}\} = 1/[j(j+1)]$ for $j \ge 1$. Both first and second inverse moments diverge. At $B = 4^J$, the ignored tail has mass $1/(J+1)$, while an ordinary/ignored split gives

$$K(B) \le 1/(J+1) + \sum_{j=1}^{J} 2^{j-J}/[j(j+1)] = 1/(J+1) + O(J^{-2}).$$

For fixed ρ and sufficiently large J, $\Psi \ge 1/(J+1)$. The upper and lower scales therefore match despite divergent full-law moments. The limit argument only needs vanishing omitted probability mass.

Why Ψ alone is insufficient. For n separated endpoint atoms $b_j = 2^{-7}4^{-j}$ with masses $w_j = c b_j^2$, normalize c and put $\rho = cn\, b_{min}^2/(1+\log n)$, $h = b_{min}/[10(1+\log n)]$. Retaining all bands is an actual optimizer. Then $K = \rho/10$ and $K/\Psi = \sqrt{(1+\log n)}$, which is unbounded. The entropy-packing term is essential even after the variational lower bound is established.

Small budgets and nonattainment. For every admissible law, $K \ge \min\{\rho, 2\sqrt{(\rho/B)}\}$, hence $K = \rho$ when $B \le 4/\rho$. The constant estimator and the lower modulus cover this regime. At fixed positive budget K is positive. The original infinite-dimensional selection infimum need not be attained: only finite-prefix lower minimizers and arbitrarily good upper decompositions are used.

8. Evidence, deliverables, and final recommendation

All checks below were independently implemented for the supplied version. Exact rational checks certify finite instances; floating-point checks are falsification tests. Neither replaces the analytic arguments for arbitrary laws, arbitrary stopping rules or compactness.

Audit file stem	Evidence retained in its result file
lower_bound_adversarial_audit	Gate stress case; 500 exact compression checks; 2000 packing laws and 732838 enumerated subsets.
midpoint_localization_audit	300 finite adaptive trees; 50048 seed-conditioned branches; overlapping components and an explicit unbounded interleaving policy.
upper_bound_adversarial_audit	Exact finite bank/race and Poisson checks; 1500 mixed-work stress cases; counterexamples to altered constructions.
finite_program_adversarial_audit	60 exact one-atom primal/dual cases; 600 flow realizations; 39040 stopping trees; 33 stored lower certificates and 9 upper policies.
infinite_limit_adversarial_audit	The analytic continuous-law counterexample; exact finite ratios and moment witnesses; 56 finite-support threshold checks.
sign_entropy_annular_audit	Weighted-overlap and saturation counterexamples; 500 exact normalization checks; constant and trimming calculations.
variational_null_work_audit	29724 pointwise optimizer checks; 500 response checks; 405 product checks; 4096 subsets in the factor-one counterexample.
main_theorem_integration_audit	5200 joint inputs, 9175 packed components, 2108 finite Poisson constructions, up to 40 levels; 120-digit near-tie resolution.

Required editorial disposition. Make the pointwise genie qualification explicit and retain the order of limits in Theorem G.3. Add the most useful interface explanations from pages 4-5. No narrowing of Theorem 3.2 or of the optimized genie theorem is supported by these audits. The ordinary-mass constant patch and other improvements are optional.

Package contents. This report; two proposed unified-diff patches; the standalone replacement theorem and counterexample in LaTeX; all eight audit scripts and result JSON files; the report builder; and a SHA-256 manifest tying them to the supplied PDF, ZIP and relevant extracted sources. Apply either patch from the extracted manuscript root. The source manuscript has not been modified.

Patch validation. Both patches pass an applicability check against the unchanged source. A scratch manuscript with both patches applied compiled successfully with the desktop LaTeX compiler. All 14 original TeX files were compared byte for byte with the supplied ZIP and remain unchanged.

Limits of the recommendation. The audits scrutinize correctness and internal consistency, not publication priority, practical polynomial-time complexity, sharp minimax constants, or extension to other observation models. The resulting assessment is favorable but provisional: the principal results withstood the stated adversarial checks, while ordinary specialist peer review remains warranted.

The correct final classification is therefore: no established fatal error in the supplied principal theorems; one explicit counterexample to an unconditional genie-convergence extension; several important proof-scope clarifications; and optional quantitative simplifications.